AHNAF SIDDIQI | email: ahnafsidd@gmail.com
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An engineer keen in cloud, network security, high traffic systems, and system design with 6+ years of experience. I love building secure products with high availability. I am also interested to learn about Cloud Networks

Tools

Experienced in: Python, C, Go, Docker, Git, Linux, Make, Nginx, PostgreSQL, MySQL, InfluxDB, Bash, X509 Certs, TCP/UDP, TLS, Gitlab, CI/CD, Flask, SQLAlchemy, HTTP(S), SNMP, MQTT, AWS, Kafka, ES, Kibana, Kubernetes, Packer

Familiar with: JS, HTML, CSS, GDB, Puppet, Redis, Grafana, CPP, MongoDB, DTLS, Ansible

Education

Singapore University of Technology and Design — Class of 2016

Bachelor of Engineering with Honors majoring in Information Systems Technology and Design.
Specialization: Security and Communications, Artificial Intelligence.

Work Experience

Visier Pte. Ltd. ~ Staff Site Reliability Engineer — Aug'25 - Now

Staff SRE in charge of cloud engineering, deploying applications products to AWS and Azure, configuring, and maintaining workforce ai agents

- Upgraded Ubunto EOL AMI/VMI to Ubuntu 24.04 with CIS hardening which is audited via usg to have hardened OS.
- Application packages are managed through puppet providing a per-application AMI/VMI.
- Application services are updated through terraform modules, which are configured for deployment via Jenkins.

Bytedance Pte. Ltd. ~ Site Reliability Engineer (Security) — Nov'22 - Aug'25

SRE in charge of 2 direct security products in a team of 25. I also manage/provision our internal Kubernetes cluster including standard sysadmin responsibilities

- Designed and implemented application metrics which captured errors which were eventually converted to P0 (user facing alerts), reducing the P0 stats by over 50%
- Metrics redesign helped to negotiate a clear SLA with our stakeholders with a enforced, data backed agreement of 99.9% which established a Runbook
- Correlated cpu, memory information which was piped into metrics with host data to provided yearly estimates of resources using a simple linear regression model.
- `strace` and `gdb` helped to identify a race condition due to $\text{futex}_{\text{wait}}$ which helped in reducing the resource application and help boost application performance.
- Stabilized apache Kafka platform which consistenly produces 76 Million messages/sec. Identified critical bottlenecks and proposed redesign of the system architecture to handle congestion.
- Provisioned Hashicorp/vault in a high availability cluster backed by Oracle KMS and a secondary vault to perform auto-unseal: https://developer.hashicorp.com/vault/docs/configuration/seal/ocikms
- Introduced blue-green/canary deployment concepts to infrastructure team with 0 downtime upgrade of applications.
- Reduced the runtime of Apache Flink job processing 20+Tb of data 10+ from 12+ hours to 4.
- Managed, reviewed, provisioned host instances, deployed application as K8s deployment/rollout/statefulset, scraped cluster metrics using prometheus agent, and collected logs using Loki.
- Actively prompted R&D to use Alpine Linux based images by providing example-led demonstration by reducing image size which saved in storage costs.
- Introduced go-mmproxy which attaches client address infront of the TCP packet header to preserve ip addresses in Layer 4 of the OSI model
- Regularly particpated in global-oncall with colleagues in 4 different timezone, while managing internal customer requriements across 4 cultures.

MicroSec Pte. Ltd. ~ Senior Software Engineer — Nov'17 - Oct'22

I created product milestones from stakeholder requirements to application implementation, lifecycle and deployment, while managing technical customer operations. I also managed the their DevOps pipeline mainly in AWS.

Role - Senior Software Engineer

- Managed companies on-premise Gitlab, scheduling regular upgrades and performing regular maintenance

- Led the deployment of 3 main products and designed the development workflow using Gitlab's runner.
- Developed a local product deployment script using bash which the developers used to simulate a full product deployment. The script is also used to demonstrate repeatable demos to internal stakeholders.
- Wrote lightweight Docker images using multi-stage build that uses alpine to significantly reduce image size.
- Developed companies Software Release Management strategies working within the Scrum methodology.
- Configured AWS IAM roles following least privlege principle for users and application system.
- Configured AWS VPC, Security Groups, Route53, ECR, EC2 and several others which is required to provision an entire deployment suites.
- Deployed and managed a production systems of 16K end-devices in a Docker Swarm for AWS

Role - Software Engineer

- Responsible for requirement gathering, analysis and translation into well-defined work description for developers to design and implement.
- Developed MicroPKI: a performant multithreaded socket server, that implements a proprietary TLS 1.3 for embedded devices in IoT strictly uses ECC Keys, specifically Ed25519
- Certificates are fine-tuned for embedded devices, with its inhouse implementation of CRL and OCSP responder
- Wrote MicroPKI protocol in C, which loads as a static library in frontend wrapper APIS including: HTTP and MQTT
- Maintained an Energy Monitoring product that uses Modbus to read localized data that is transferred near real-time via MQTT that is batched process and stored in InfluxDB.
- Led the design and implementation of RESTful microservices using python that supports in-house implementations of ACMEv2, EST, and SCEP following their respective RFCs: 8555, 7030, 8894.
- Introduced the automation of backend e2e testing with unit test at microservices level and integration test at a separate product level.
- Implement simulation testbeds that replicates a CANBUS protocol scenario using Docker.
- Identified a missing parser in Microsofts Library for SCEP protocol after parsing the ASN.1 certificate data structure.
- Led Scrum Master duties for two teams of 10+ individuals.
- Evangelize the Scrum philosophy with the development team to promote code reviews and collaboration.
- Increased team delivery yield by a factor of 1.5, rotating or setting expectations after various 1-1 meetings.
- Negotiated product deliverable timeline per sprint between different internal stakeholders and external clients
- Mentored several interns to full time employees following companies guidelines.

Singapore University of Technology and Design ~ Research Assistant Nov'16 - Oct'17

Implemented a concurrent Common Industrial Protocol (CIP), a realtime network protocol at Secure Water Treatment (SWaT) testbed in SUTD, that was used in simulation, mutliple research projects, and as proxies for developing attack scenarios for competitions.

- Packet Sniffing and Reconstruction of high volume CIP packets for both TCP and UDP.
- Designed CRUD HTTPS API to handle application requests using using TDD and SOLID principles of OOP.

Publications

Patent: MicroSec's patent for securing communications in Embedded Systems that uses a proprietary PKI based protocol

Vishram Mishra, A. Siddiqi et al (2022). Method and Architecture for Securing and Managing Networks of Embedded Systems with Optimised Public Key Infrastructure (20220006652-A1). U.S. Patent and Trademark Office. Pending

Paper: Detailed network segment, protocol, and service analysis of EPIC Industrial testbed

A. Siddiqi, N. O. Tippenhauer, D. Mashima, and B. Chen, "On practical threat scenario testing in an electric power ics testbed," in Proceedings of the cyber-physical system security workshop (cpss), co-located with asiaccs, 2018. https://doi.org/10.1145/3198458.3198461

Certifications

Certified DevSecOps Professional (CDP) - May 2022

- Issuer: Practical DevSecOps
- Badge Link